

First Individual Assessment

 Published

 Assign To

 Edit



COMS W4156 Advanced Software Engineering (ASE)

First Individual Assessment Fall 2023

This is an individual assignment. You are NOT permitted to post, discuss or work together on the assessment questions with other students in the class or with anyone else other than the teaching staff. Your answers can be as long or as short as you want, but a long meandering answer that incidentally mentions something relevant will receive less credit than a short high-quality answer. Submit your answers all in a single document in response to this assignment on courseworks. You can submit as many times as you want until the deadline. Submissions via email, edstem, or any other means besides courseworks will not be graded and will receive zero credit.

Academic Honesty policy: All submissions will be automatically analyzed by turnitin.com for plagiarism/collusion. Plagiarism or collusion will result in zero credit on the assessment and referral to the relevant disciplinary committee.

Late policy: Submissions posted one day late (which includes everything from 1 minute late to 23 hours 59 minutes late) will receive maximum 75% credit; two days late maximum 50% credit; three days late maximum 25% credit; four days late 0 credit.

Maximum total 100 points. There is no credit for part 0, but 0A is required in the sense that we will not grade the graded questions if part 0A is missing or incomplete and you will receive zero credit for the assessment. Part 0B is optional and will not affect your grade.

Ungraded questions: part 0A is required and part 0B is optional.

part 0A (required): You have 100 total points to allocate among the members of your team. **Assign some number of points (possibly 0) to each member of your team based on what you believe was that student's relative participation in the team project thus far, and *explain your scores*.** Make sure the total is exactly 100. Remember to include yourself! For allocating points, consider level of effort and contributions for anything relevant to the team project. Note equal participation is not the same thing as equal lines of code produced. Please do not predict the future, consider only the contributions to the team project as of the date of writing. It is very important that you explain the scores you give to each student, including yourself: *If you do not provide meaningful explanations, then your response will be considered incomplete and you will receive 0 overall on the assessment.*

For example, in a team where all team members participated more or less equally, the scores would be: Cora 25, Kaedi 25, Kaylah 25, Nina 25, and your explanation would describe who did which part(s) of the team project effort thus far.

In a team with a substantial disparity in contributions, however, the scores might be: Hawkeye 35, Captain Marvel 35, Hulk 25, Thor 5, and your explanation might describe Hawkeye and Captain Marvel as doing most of the work, and Hulk helped, but Thor rarely showed up for team meetings and did not seem interested in the project. Again, your explanation should describe who did what, or who did not do what, as the case may be.

part 0B (optional): Is there anything that you think should have been covered already but wasn't? Is there anything that was covered unnecessarily or should have been put off until later? Is there anything that should have been presented in a different way? Is there anything else you would change about the course? Please explain. Suggestions for the remainder of this course will be greatly appreciated. In addition to submitting comments and suggestions here as part of your first assessment, you can also post anonymously on Edstem. (And please also submit the SEAS evaluation.)

continue to next page

Graded questions: There are two graded questions, part 1 and part 2.

Part 1 (50 points): The company that employs your team originally planned to offer your service “for free”, and you already have several external clients, but is now rethinking that business model. Higher management has decided to monetize your service by introducing advertising. In particular, your team has been tasked to extend your API with special entry points just for advertiser clients that are not available to other clients, which those advertiser clients will use to upload ads. Further, you also need to augment the data your regular API sends to regular clients to include one randomly selected ad embedded in the regular data. However, regular clients can ‘opt-out’ of receiving these ads by also paying money, so your service needs to track which clients have opted-out and which have not, and only send the embedded ads to clients that have not paid.

Assume all the monetary aspects including opting out are handled offline, not part of your service’s API, and your service is administratively provided with information about the advertiser clients and the opt-out clients, where any and all other clients are not-opt-out. Note the set of advertiser and opt-out clients may change over time. Also assume the ads are well-formed in whatever format makes sense for your data, such as text or images that your service could add to the beginning, middle, or end of the regular data returned to clients. It is not necessary to look into how real-world advertising actually works, make something up.

Include in your answer a link to your team project github repository but do not change anything in your team’s github repository in response to this question. Make sure your repo is public or, if private, the instructor and all IAs have collaborator access.

You do not need to write any real code for this assignment, but pseudo-code may be useful as part of your answers.

Part 1.A (20 points): Invent some new API entry points specifically for paid advertisers, explain what they do, describe how they leverage your service’s existing code, and discuss in detail how your service

will ensure that these entry points are used only by paid advertiser clients and not by any other clients.

Part 1.B (20 points): Describe in detail how your service will make sure to serve an ad to every non-opt-out client with every response that sends data to that client, but never send any ads to opt-out clients or advertiser clients. If you would need to change your regular client API in some way, besides including an ad as part of each data response, explain how and why.

Part 1.C (10 points): Discuss in detail what you would need to do to test this new functionality of your service and also test that the new functionality does not break any of the original functionality.

Part 2 (50 points): Imagine that your team project was required to develop a *library* instead of a *service*, with the intent that the library could be used by other developers as part of their own application or service and run inside the same operating system process rather than being accessed over the network. You may need to change the functionality of your system in various ways. Most notably there needs to be a separate instance of the library inside every client process rather than a single instance of the service running standalone in its own process. (For this question, do not include the advertising functionality introduced above, only consider your system's actual functionality.)

Include in your answer a link to your team project github repository but do not change anything in your team's github repository in response to this question. Make sure your repo is public or, if private, the instructor and all IAs have collaborator access.

You do not need to write any real code for this assignment, but pseudo-code may be useful as part of your answers.

part 2a (20 points): What specific changes would be needed in your system to provide similar functionality as a library instead of a service? Explain how those changes should be reflected in your API entry points. If any of the entry points would no longer be applicable and should be removed, explain why. If any new entry points would be needed to retain similar functionality as a library, explain what they would do and how they would work. Make sure to consider all of your system's existing entry points.

part 2b (20 points): What specific changes would be needed in the implementation of the persistent storage aspect of your system and why? What specific changes would be needed in the implementation of the multi-client aspect of your system now that there is only one client?

part 2c (10 points): Discuss in detail what you would need to do to test this new library version of your project.

This is the end of the assessment.

Submitting a file upload

File Types pdf, doc, docx, and txt

Due	For	Available from	Until
Oct 30, 2023	Everyone	Oct 27, 2023 at 12:01am	Nov 3, 2023 at 11:59pm

First Assessment
Rubric

You've already rated students with this rubric. Any major changes could affect their assessment results.

Criteria	Ratings	Pts
1A) Invent some new API entry points specifically for paid advertisers, explain what they do, describe how they leverage your service's existing code, and discuss in detail how your service will ensure that these entry points are used only by paid advertiser clients and not by any other clients. (max 20pts) Define your new API endpoint(s) [10pts] - Signature [2pts] - Return [2pts] - Explanation of what it does [6pts] Any form of ensuring that the advertisers are "valid advertisers" [5pts] - Authentication (or equivalent method) [1pts] - Describe how the proposed method can correctly distinguish between advertisers and non-advertisers [3 pts] - How existing code is leveraged (full credit if not applicable) [1pts] Storing the ad in a persistent storage [5pts] - Mentioned persistent storage (but allow any accepted ad format) [4pts] - How existing code is leveraged (full credit if not applicable) [1pt]	This area will be used by the assessor to leave comments related to this criterion.	20 pts
1B) Describe in detail how your service will make sure to serve an ad to every non-opt-out client with every response that sends data to that client, but never send any ads to opt-out clients or advertiser clients. (max 20pts) Change response API(s) to include an ad [10 pts] - Proposed change would cover all possible responses [1 pts] - Describe the modifications made to the return type of API endpoints (full credit if not applicable) [3 pts] - Describe the difference in the response of the API for an opt-out vs non opt-out client [3 pts] - Should be a generic, scalable change (i.e., without having to change every small part of the response for both opt-out/non opt-out clients) [3 pts] Ensures only non-opted out client will receive the ad [10 pts] - Proposed change would cover all possible responses [1 pts]	This area will be used by the assessor to leave comments related to this criterion.	20 pts

Criteria	Ratings	Pts
- A method to check the identity of the API caller [3 pts] - - E.g., check the role of the client using your database - Describe how only non-opted out clients will be affected [3 pts] - Should be a generic, scalable method [3 pts]		
1C) Discuss in detail what you would need to do to test this new functionality of your service and also test that the new functionality does not break any of the original functionality (max 10pts) For every APIs implemented (including the new API), do: API testings need to include/cover [10 pts] - Test for the advertiser role (where applicable) [2 pts] - Test for the opt-out client role (where applicable) [2 pts] - Test for non-opt out client role (where applicable) [2 pts] - Edge cases are included and justified [1 pt] - All regular API calls are covered [3 pts] - - E.g., changes did not break any original functionalities irrelevant to the advertising	This area will be used by the assessor to leave comments related to this criterion.	10 pts
2A) What specific changes would be needed in your system to provide similar functionality as a library instead of a service? Explain how those changes should be reflected in your API entry points. (max 20pts) Discussion includes changing the service to be able to serve the developer client/process independently instead of serving multiple direct clients at once. [5 pts] Changing existing API functions to library functions [9 pts] - Functions are no longer HTTP/web [5 pts] - - change signature if applicable - - change return value, if applicable - Proposed changes align with how a library is typical used [3 pts] - - E.g., library should not just act as wrapper around service - - Ignore anything to do with persistent storage because that is in 2b) - Coverage of all entry points [1 pt] Adding new/removing existing code [6 pts]: Moving away from any web-framework related dependencies for auth,	This area will be used by the assessor to leave comments related to this criterion.	20 pts

Criteria	Ratings	Pts
session management, routing, error logging, etc. [3 pts] Any changes required for their logic to work as an embedded library rather than over HTTP (full credit if not applicable) [3 pts]		
2B) What specific changes would be needed in the implementation of the persistent storage aspect of your system and why? What specific changes would be needed in the implementation of the multi-client aspect of your system now that there is only one client? (max 20pts) Ensure library is not using a pre-specified, central storage system [14 pts] - Mentioned that persistent storage will need be configured and managed by the direct client [6 pts] - Describe how clients could set up their own database. [4 pts] - Describe how clients could configure their database now that they own it. [4 pts] - - (e.g. the ability for a library user to select their own supported drivers/connections, etc) Include any changes to your existing code/logic to handle the new persistent storage system [6 pts] - Mention any changes to the logic now that each client would have their own persistent storage system (full credit if not applicable) [3 pts] - Ensure that your library can work independently for each client [3 pts] - - I.e., your library will not be affected by multiple client processes running concurrently	This area will be used by the assessor to leave comments related to this criterion.	20 pts
2C) Discuss in detail what you would need to do to test this new library version of your project. (max 10pts) Mentions tests on all new/modified APIs (testing 2a) [5 pts] - Consider test setup, tear down, and/or fixture changes if applicable. (1 pts) [1 pt] - Explain non-database related changes required due to changing the service to a library [3 pts] - - E.g., signature and return - Coverage of all entry points [1 pt] Considers how storage changes affects your testing (testing 2b) [5 pts]:	This area will be used by the assessor to leave comments related to this criterion.	10 pts

Criteria	Ratings	Pts
- Consider test setup, tear down, and/or fixture changes if applicable. [1pts] - Explain database-related changes required due to changing the service to a library [3 pts] - - E.g., reading and writing from the client's persistent storage system - Any logic that does not deal with storage directly should be unaffected (full credit if not applicable) [1 pt]		
Total Points: 100		